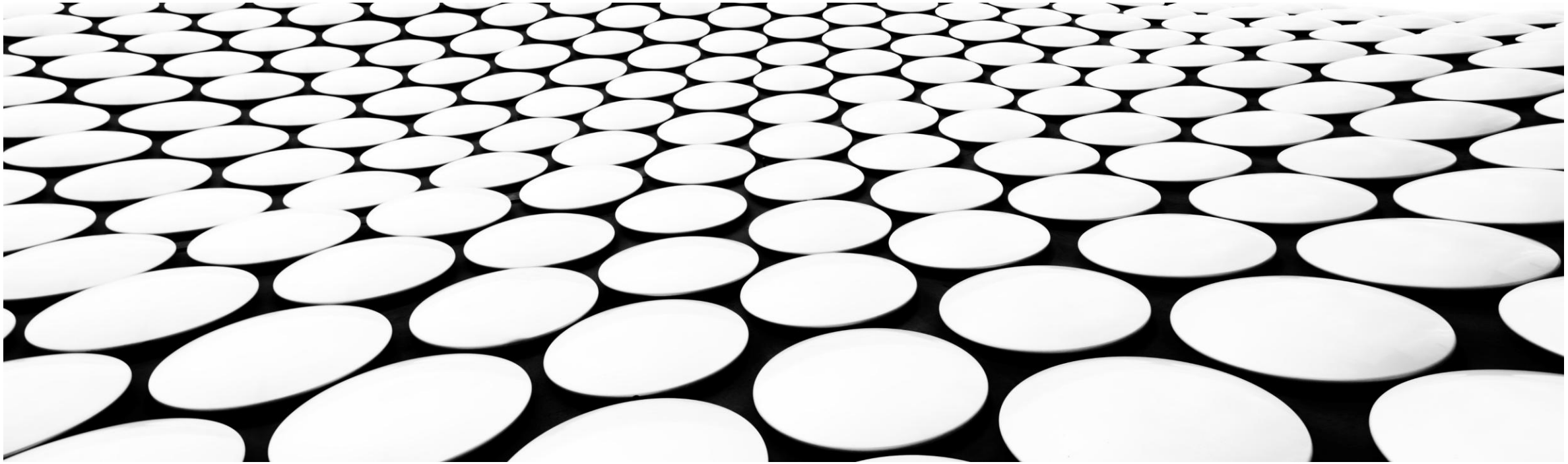
INTER NETWORKING DEVICES



CONTENTS

- Layer 1 connections- Repeater, Hubs
- Layer 2 connections- Bridges, Switches
- Layer 3 connections- Switches, Routers, Gateways

CONNECTING DEVICES

NETWORKING

Repeaters



Receive, regenerate and retransmit signals to extend network range.

Bridges



Connect two network segments and filter traffic based on MAC addresses.

Hub



Connect multiple devices in a network. Broadcasts data to all ports.

Switches



Connect different devices and forward data to the correct destination using MAC addresses.

INTERNET-WORKING

Routers



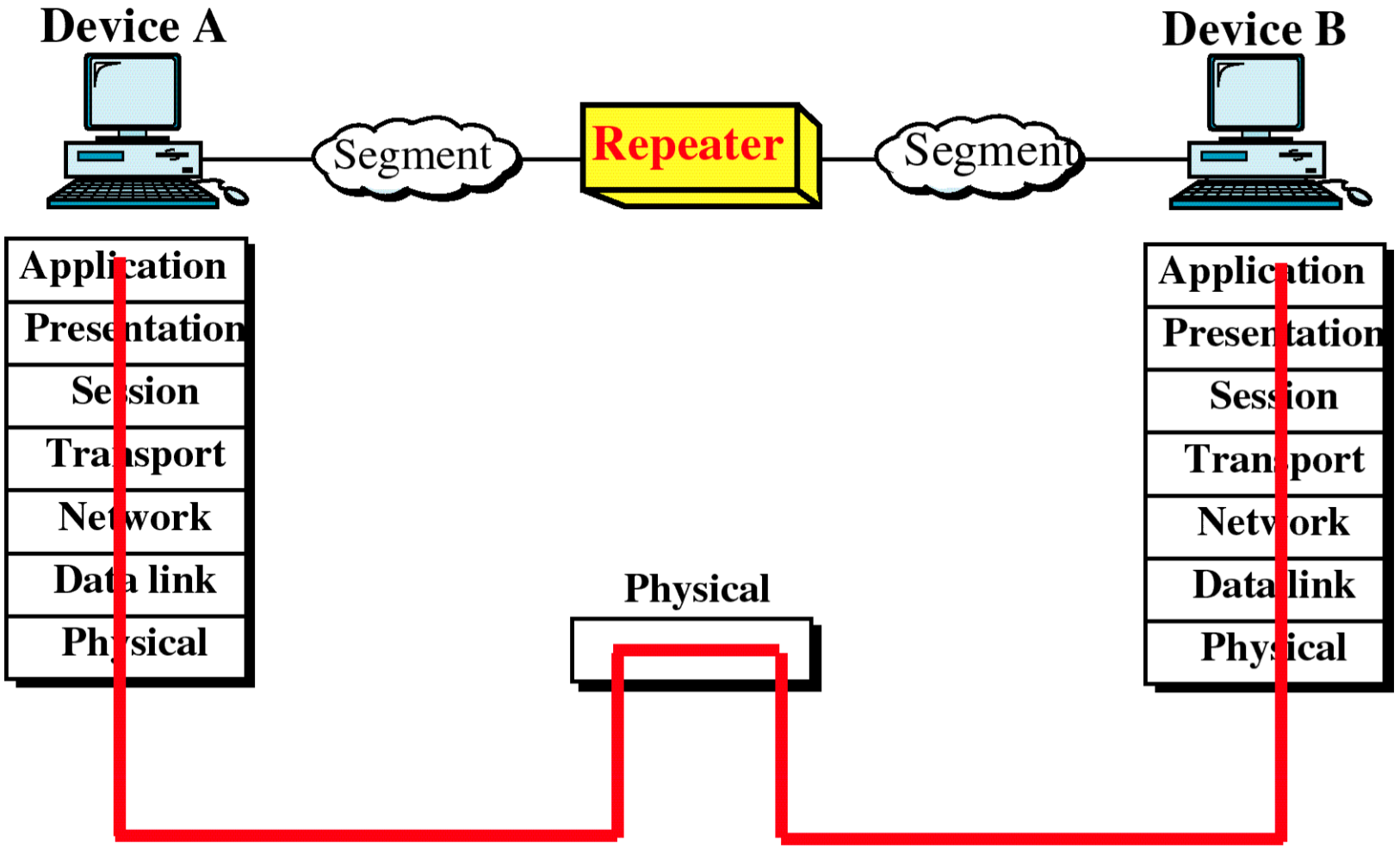
Connect different networks and route data packets to the best path.

Gateways



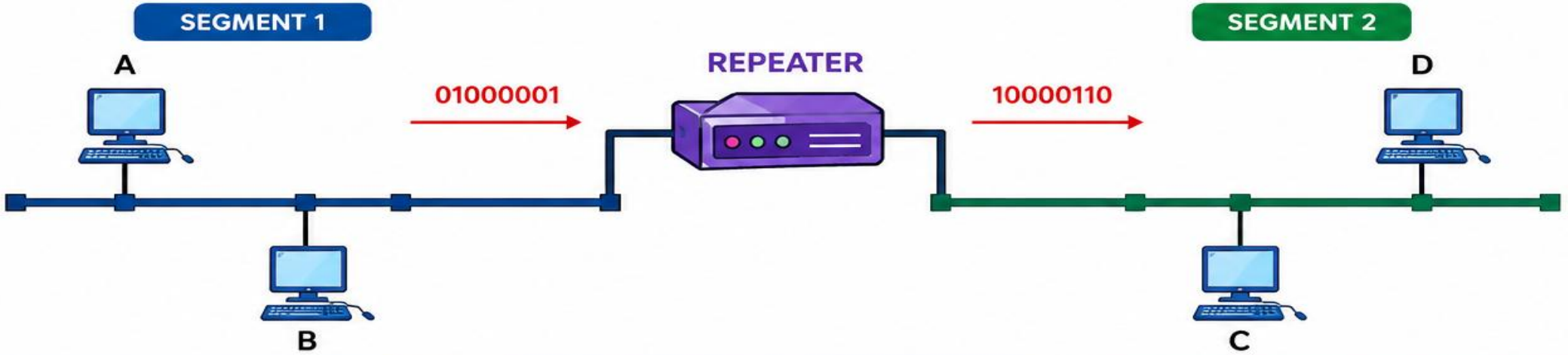
Connect different network protocols or systems and enable communication between them.

Repeater and OSI Model



REPEATERS IN COMPUTER NETWORKS

A **Repeater** is a physical layer device used to **regenerate, amplify** and **retransmit** signals so that data can travel longer distances without distortion or loss.



WHAT DOES A REPEATER DO?

- Receives weak or distorted signals
- Regenerates and amplifies the signal
- Re-transmits the signal to the next segment
- Operates at the Physical Layer (Layer 1)
- Does not understand the data content

KEY FEATURES

- Extends the range of a network
- Regenerates both digital and analog signals
- Does not filter or correct errors
- Simple and inexpensive device
- Used with bus, star, or linear topologies

WHERE ARE REPEATERS USED?

- In larger networks to extend cable length
- In bus topologies (e.g., Ethernet 10Base2, 10Base5)
- In noisy environments to boost signal strength
- Between two network segments

HOW IT WORKS



ANALOGY



IMPORTANT NOTES

- ★ A repeater only extends the distance.
- ★ It does not increase bandwidth.
- ★ Too many repeaters can increase delay.
- ★ Modern networks use switches and routers instead of repeaters.

SUMMARY: A repeater is like a signal booster that helps data travel farther by cleaning and strengthening the signal without changing the actual data.

Function of a Repeater



(a) Right-to-left transmission.



(b) Left-to-right transmission.



HUB

A hub is a basic networking device that connects multiple devices in a network.

What is a Hub?

A hub is a multi-port repeater.

It connects multiple devices and broadcasts incoming data to all ports.

FUNCTION

- Operates at the Physical Layer (Layer 1)
- Receives data from one device and broadcasts it to all other connected devices.
- Does not inspect or filter data.
- All ports share the same bandwidth.
- Available in two types:
  - **Passive Hub** – only concentrates connections.
  - **Active Hub** – concentrates connections and regenerates (boosts) signals.

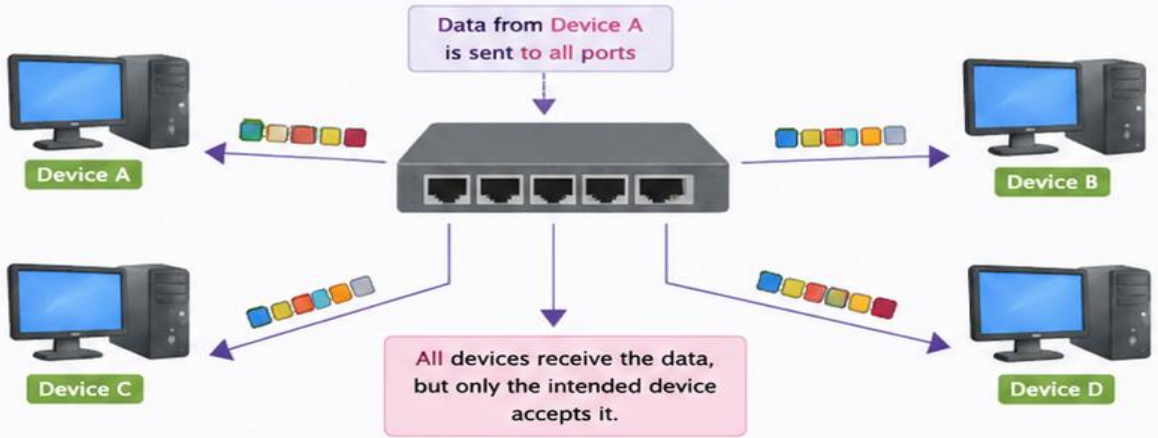
APPLICATIONS

- Connecting small networks in homes or offices.
- Used in environments where cost is a major concern.
- Suitable for temporary setups or testing networks.
- Still used in some legacy or low-speed networks.

LIMITATIONS

- All devices share the same bandwidth, leading to collisions and low performance.
- Operates in half-duplex mode only.
- No security features.
- Does not filter or manage data traffic.
- Not suitable for large or high-traffic networks.

HOW A HUB WORKS



EXAMPLE TO IDENTIFY

Hubs usually have multiple RJ45 ports and LED indicators. They do not have any MAC address table and treat all ports equally.



Key Point

Hubs are simple and inexpensive devices used to connect multiple devices, but they are now mostly replaced by switches in modern networks due to performance and security limitations.



LAYER 2 SWITCHES

Smart Switching at the Data Link Layer

A Layer 2 Switch operates at OSI Layer 2 (Data Link Layer) and uses MAC addresses to forward data within a LAN efficiently.

KEY BENEFITS

- High Performance & Low Latency
- Reduces Network Congestion
- Improves Security (VLANs, Port Security)
- Scalable and Cost-Effective
- Easy to Deploy and Manage

KEY FEATURES

MAC Address Learning
Learns MAC addresses from source devices and builds a MAC address table.

MAC Address Forwarding
Forwards frames only to the port where the destination MAC is located.

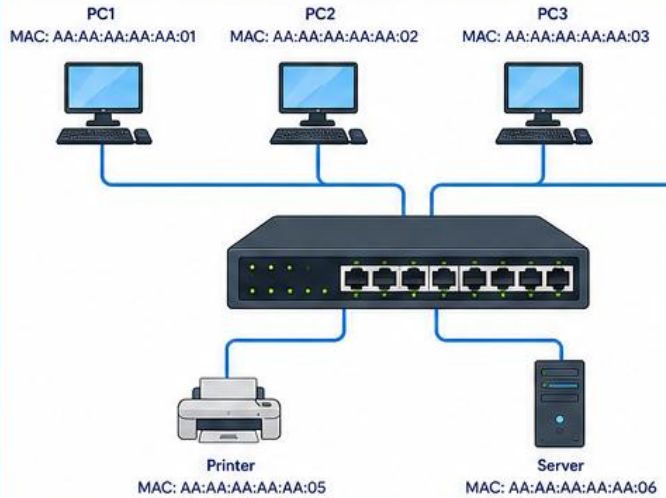
Filtering
Filters and drops frames that are not meant for any device on the network.

Reduces Broadcast Domain
Each port is its own collision domain, improving network efficiency.

Supports VLANs (802.1Q)
Segment networks logically using VLANs for better management and security.

Spanning Tree Protocol (STP)
Prevents loops and ensures redundancy without broadcast storms.

HOW LAYER 2 SWITCH WORKS



MAC ADDRESS TABLE (EXAMPLE)

MAC Address	Port
AA:AA:AA:AA:AA:01	1
AA:AA:AA:AA:AA:02	2
AA:AA:AA:AA:AA:03	3
AA:AA:AA:AA:AA:04	4
AA:AA:AA:AA:AA:05	5
AA:AA:AA:AA:AA:06	6

FRAME FORWARDING PROCESS

1. **Frame Received on Port** — Frame comes in from any port.
2. **Check MAC Table** — Switch checks the destination MAC address in its table.
3. **Forward or Filter** — If found, forward to that port. If not found, flood to all ports (except incoming port).
4. **Update Table** — Source MAC address is added to the MAC table.

LAYER 2 SWITCH SPECIFICATIONS

Operating Layer	Layer 2 (Data Link Layer)
Address Used	MAC Address
Forwarding Method	Store-and-Forward
Switching Capacity	High (Non-blocking)
VLAN Support	Yes (802.1Q)
STP Support	Yes (802.1D / RSTP / MSTP)
Broadcast Domain	Same for all ports (unless VLANs are used)
Collision Domain	Separate per port
Typical Use	LANs, Office Networks, Campus Networks

COMMON USE CASES

Enterprise Networks · Educational Institutions · Healthcare Networks · Surveillance Networks · VoIP Networks

TYPES OF LAYER 2 SWITCHES



- **Unmanaged Switch** — Plug and play, no configuration
- **Smart Managed Switch** — Basic management features
- **Managed Switch** — Full configuration and monitoring
- **PoE Switch** — Provides power to devices over Ethernet

IMPORTANT NOTES

- Layer 2 switches do not route packets between networks.
- They work based on MAC addresses, not IP addresses.
- Use VLANs to segment networks logically.
- Ideal for connecting devices within the same network.

SUMMARY: Layer 2 Switches are essential for efficient data communication in LANs. They improve performance, reduce collisions, and provide advanced features like VLANs and STP for better network control and security.

LAYER 3 SWITCHES

High Performance Switching + Routing Intelligence



A Layer 3 Switch operates at OSI Layer 3 (Network Layer) and combines the high-speed switching of Layer 2 with the routing capabilities of a router.

KEY FEATURES

Routing Capability
Routes traffic between VLANs and different IP subnets at wire speed.

High Performance
Hardware-based routing and switching for low latency and high throughput.

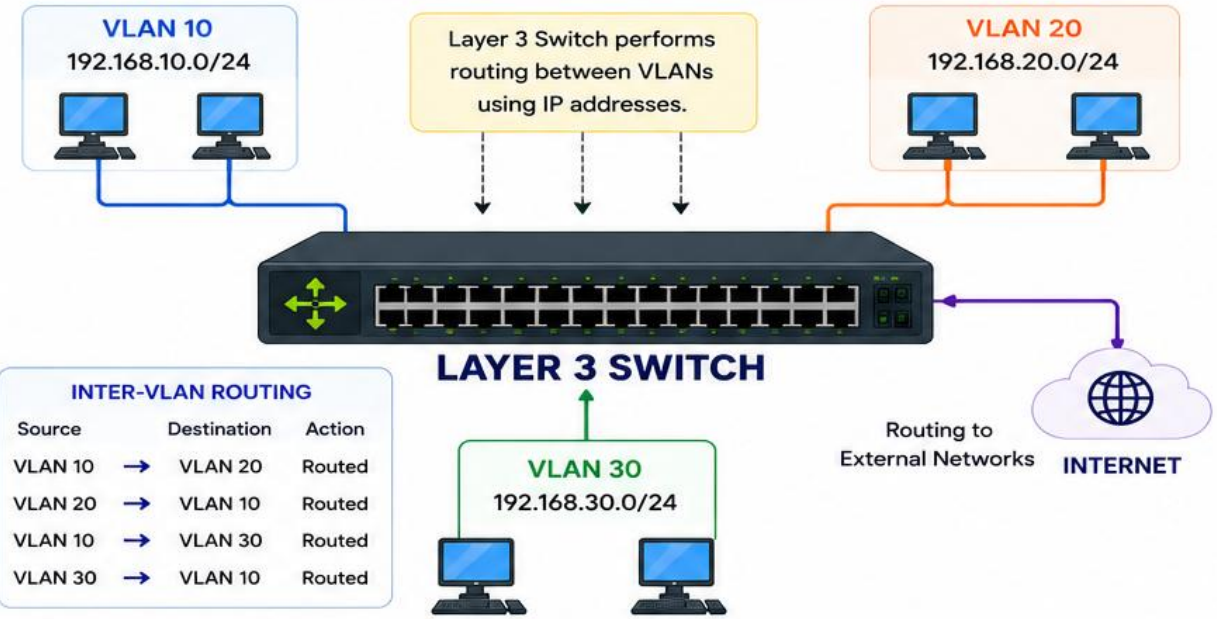
VLAN Support
Supports inter-VLAN routing and VLAN interfaces (SVI).

Access Control & Security
Supports ACLs, DHCP relay, IP source guard, and more.

Scalability
Designed for large networks with multiple VLANs and subnets.

Advanced Management
Supports QoS, Policy-Based Routing, SNMP, and more.

HOW LAYER 3 SWITCH WORKS



INTER-VLAN ROUTING

Source		Destination	Action
VLAN 10	→	VLAN 20	Routed
VLAN 20	→	VLAN 10	Routed
VLAN 10	→	VLAN 30	Routed
VLAN 30	→	VLAN 10	Routed

WORKING PROCESS

1. **Packet Arrives** — Packet arrives at the Layer 3 Switch on a port.
2. **Check Destination** — Switch checks the destination IP address.
3. **Routing Decision** — If the destination is in another subnet/VLAN, routing is required.
4. **Route & Forward** — Switch routes the packet to the best path and forwards to next hop.
5. **Packet Delivered** — Packet reaches the destination device in the target VLAN.

SUMMARY: Layer 3 Switches provide the speed of Layer 2 switching with the intelligence of Layer 3 routing, enabling efficient, secure, and scalable network communication.

LAYER 3 SWITCHES

Benefits, Use Cases & Comparison

BENEFITS

Reduces the need for a separate router
Combines routing and switching in one device.

Faster inter-VLAN routing
Hardware-based routing delivers high speed and low latency.

Simplifies network design
Centralized routing and switching simplifies management.

Improves network performance
Efficient traffic flow and reduced bottlenecks enhance performance.

Enhanced security
Supports ACLs, IP filtering, QoS and other advanced security features.

Cost-effective
Cut down hardware costs, power usage and operational costs.

COMMON USE CASES

Enterprise Networks
Connect multiple departments and VLANs efficiently with high-speed routing.

Campus Networks
High-performance inter-VLAN routing across multiple buildings and floors.

Data Centers
Fast east-west traffic, server segmentation and high availability.

Service Provider Networks
Customer VLANs and subnet routing with high scalability.

Cloud & Hybrid Networks
Connect on-premise networks to cloud and external networks securely.

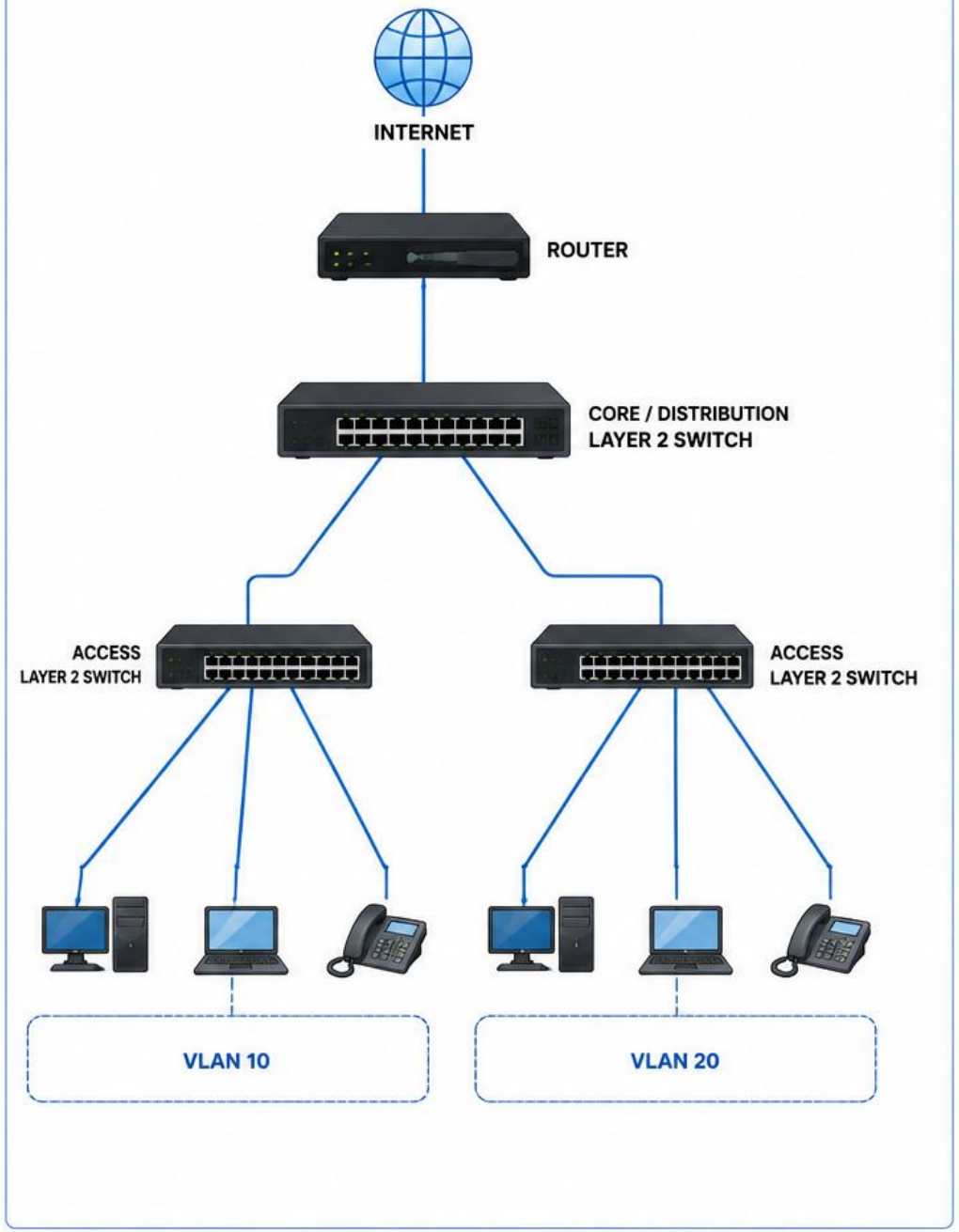
LAYER 2 SWITCH vs LAYER 3 SWITCH

FEATURE	LAYER 2 SWITCH	LAYER 3 SWITCH
OSI Layer	Layer 2 (Data Link)	Layer 3 (Network)
Primary Function	Switching	**Switching + Routing**
Routing Capability	No	Yes (Inter-VLAN & Inter-Subnet Routing)
IP Address Awareness	No	**Yes**
Inter-VLAN Routing	No (Requires External Router)	Yes (Built-in)
Performance	High for Local Switching	**Very High** (Switching + Routing in Hardware)
Ports	Many (24 / 48+)	Many (24 / 48+)
Broadcast Domain	One per VLAN	**Can Break Broadcast Domains** (Routes Traffic)
Typical Use	Small / Simple LANs	Enterprise / Campus / Data Centers
Cost	Lower	**Moderate to High**

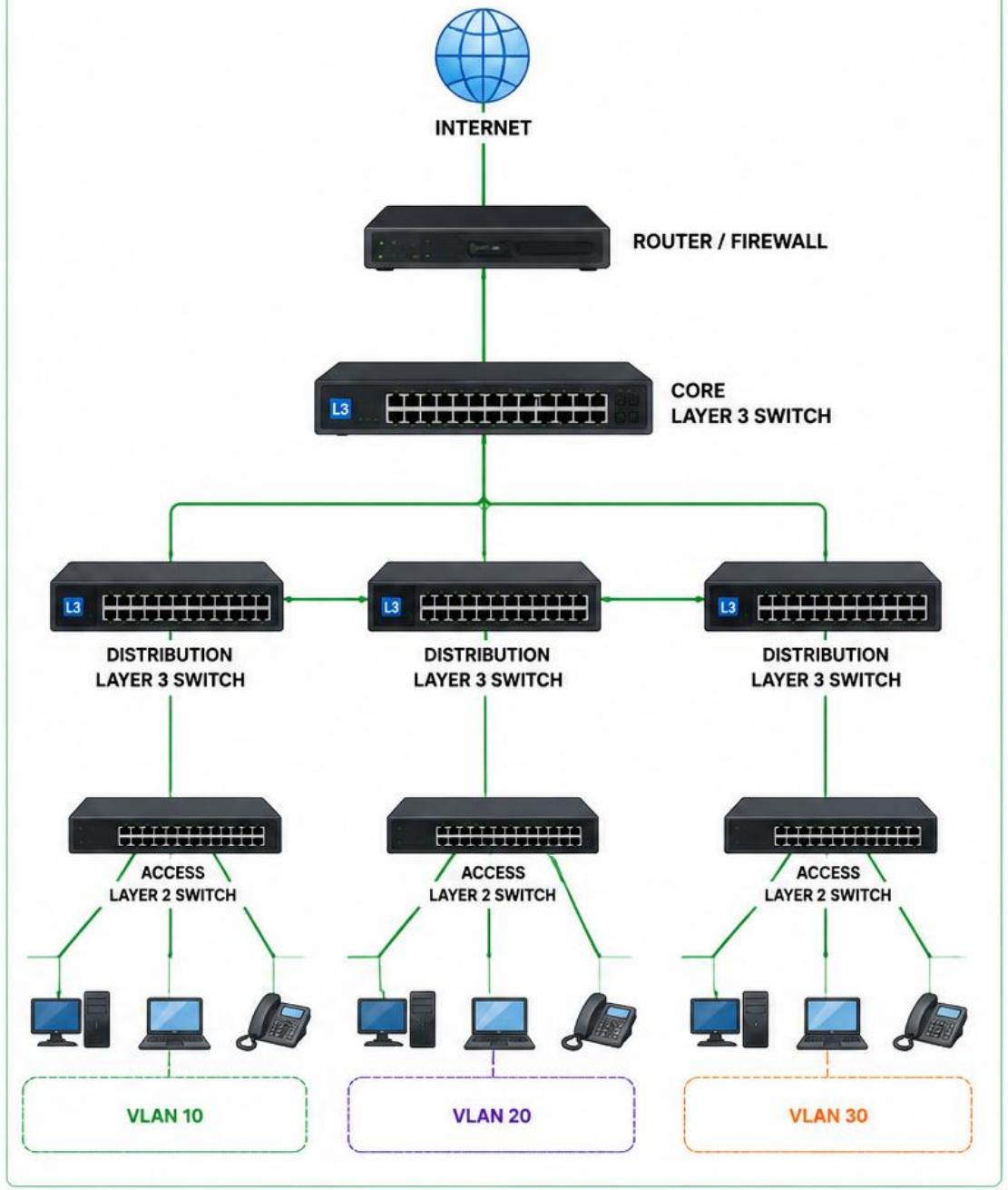
SUMMARY: Layer 3 Switches combine high-performance switching with routing intelligence, making them ideal for modern networks that require speed, scalability and efficiency.



LAYER 2 SWITCH NETWORK TOPOLOGY



LAYER 3 SWITCH NETWORK TOPOLOGY

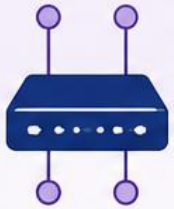


BRIDGES

CONNECTING NETWORKS INTELLIGENTLY

WHAT IS A BRIDGE?

A bridge is a networking device that connects two or more network segments at the Data Link Layer (Layer 2) of the OSI model.



A bridge divides a large network into smaller segments to improve performance and reduce congestion.

BRIDGE DEVICE



Typical Bridge with multiple ports

FUNCTION

- Connects two or more network segments.
- Operates at the Data Link Layer (Layer 2).
- Learns MAC addresses from incoming frames.
- Forwards data only to the necessary segment.
- Filters unnecessary traffic based on MAC addresses.
- Reduces network congestion and improves performance.

APPLICATIONS

- Connecting similar networks (LAN to LAN).
- Segmenting a large network into smaller parts.
- Extending the range of a network.
- Used in campus networks, office networks, and data centers.
- Helpful in reducing traffic and improving network efficiency.

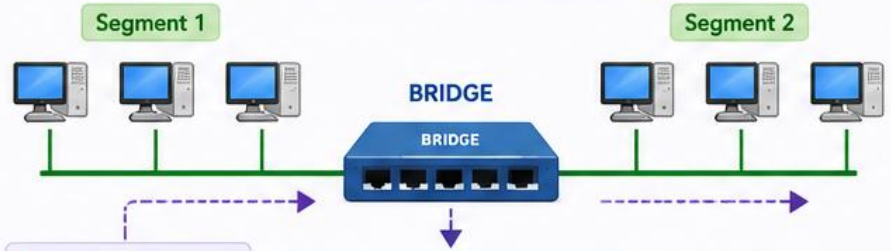
LIMITATIONS

- Works only with the same type of network.
- Does not connect dissimilar networks (e.g., Ethernet to Token Ring).
- Cannot route data between different network protocols.
- Limited to Layer 2 – does not understand IP addresses.
- Broadcasts and unknown frames are forwarded to all segments.

KEY FEATURES

- Operates at Layer 2 (Data Link Layer).
- Uses MAC addresses for forwarding.
- Maintains a MAC address table.
- Divides network into collision domains (but not broadcast domains).
- Transparent to users.
- Plug-and-play device.

HOW A BRIDGE WORKS



1. Device in Segment 1 sends data to Device in Segment 2.
2. Bridge checks the MAC address table and forwards the frame only to the correct segment.
3. Data is delivered to the destination in Segment 2. Other devices are not affected.

EXAMPLE

Without Bridge



All devices share same bandwidth (high collisions)

With Bridge

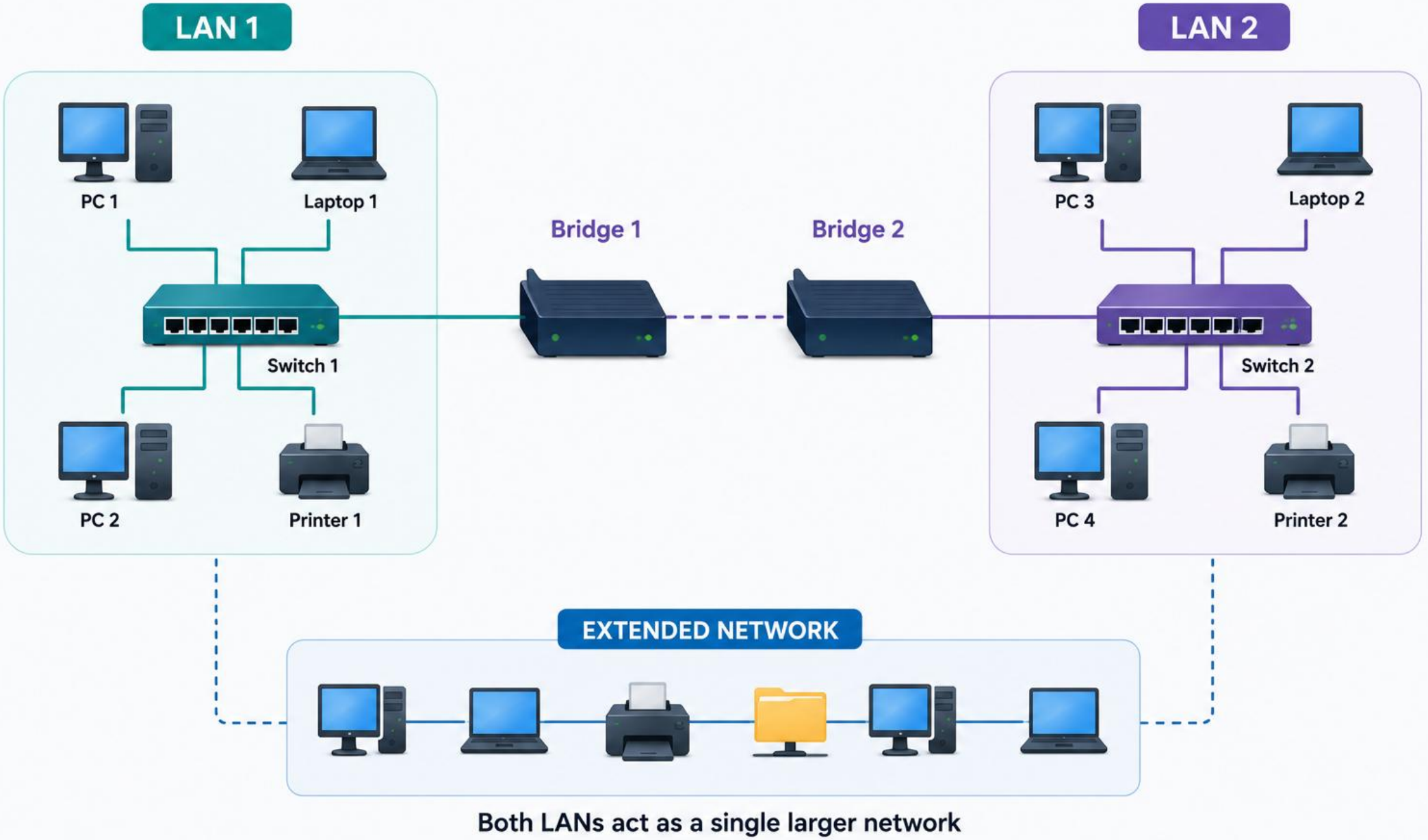


Traffic is filtered. Better performance & fewer collisions.

KEY POINTS

- Bridge = Smart Connector at Layer 2
- Learns, Filters, Forwards
- Reduces Traffic & Improves Network Performance

BRIDGES IN A NETWORK



Both LANs act as a single larger network



ROUTERS

Smart Device that Connects Different Networks



KEY FEATURES

- Operates at Network Layer (Layer 3) of the OSI model
- Connects different networks (e.g., LAN to WAN)
- Uses IP addresses to find the best path for data
- Supports routing protocols to exchange route information
- Separates broadcast domains (reduces unnecessary traffic)
- Provides security features like firewall and NAT

FUNCTIONS

Routing
Chooses the best path and forwards data to the destination network.

Path Selection
Selects the most efficient route based on traffic and network conditions.

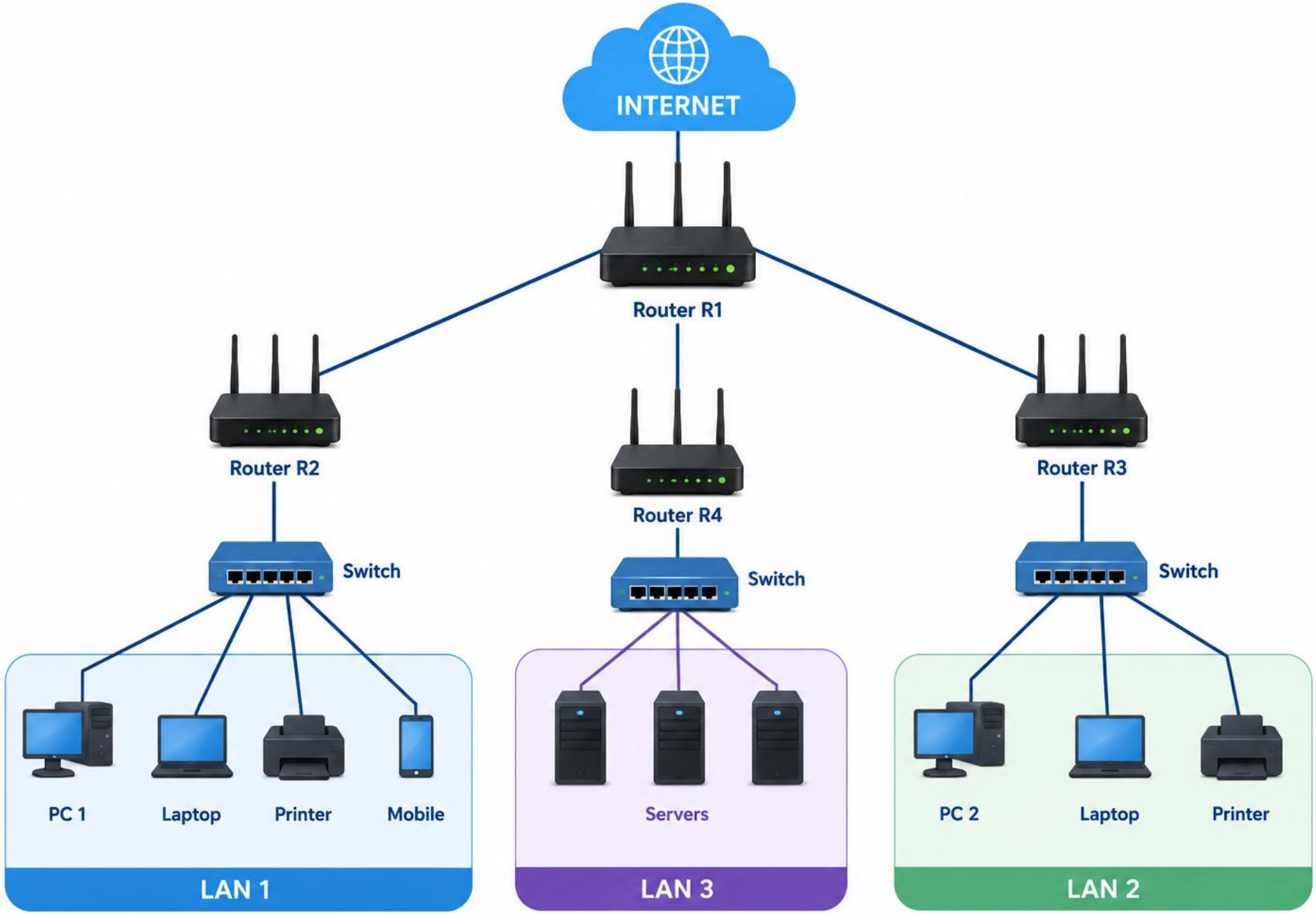
Traffic Management
Manages and directs data traffic to reduce congestion.

Network Security
Protects the network using firewall, NAT, and access control.

APPLICATIONS

- Connecting multiple LANs in an organization
- Connecting LAN to the Internet (WAN connection)
- Used in home networks with broadband routers
- Used in data centers and cloud networks
- Used by ISPs to connect users to the Internet

Router = Connects Different Networks + Finds Best Path + Forwards Data Efficiently


INTERNET
Router R1
Router R2
Router R4
Router R3
Switch
Switch
Switch
PC 1
Laptop
Printer
Mobile
Servers
PC 2
Laptop
Printer
LAN 1
LAN 3
LAN 2



GATEWAY

Connects Different Networks Using Different Protocols



FUNCTIONS

- Connects different networks
- Translates different protocols
- Converts data formats
- Provides access between dissimilar systems

KEY FEATURES

- Works at multiple layers
- Handles different protocols & formats
- Ensures compatibility between networks
- Acts as a translator between systems

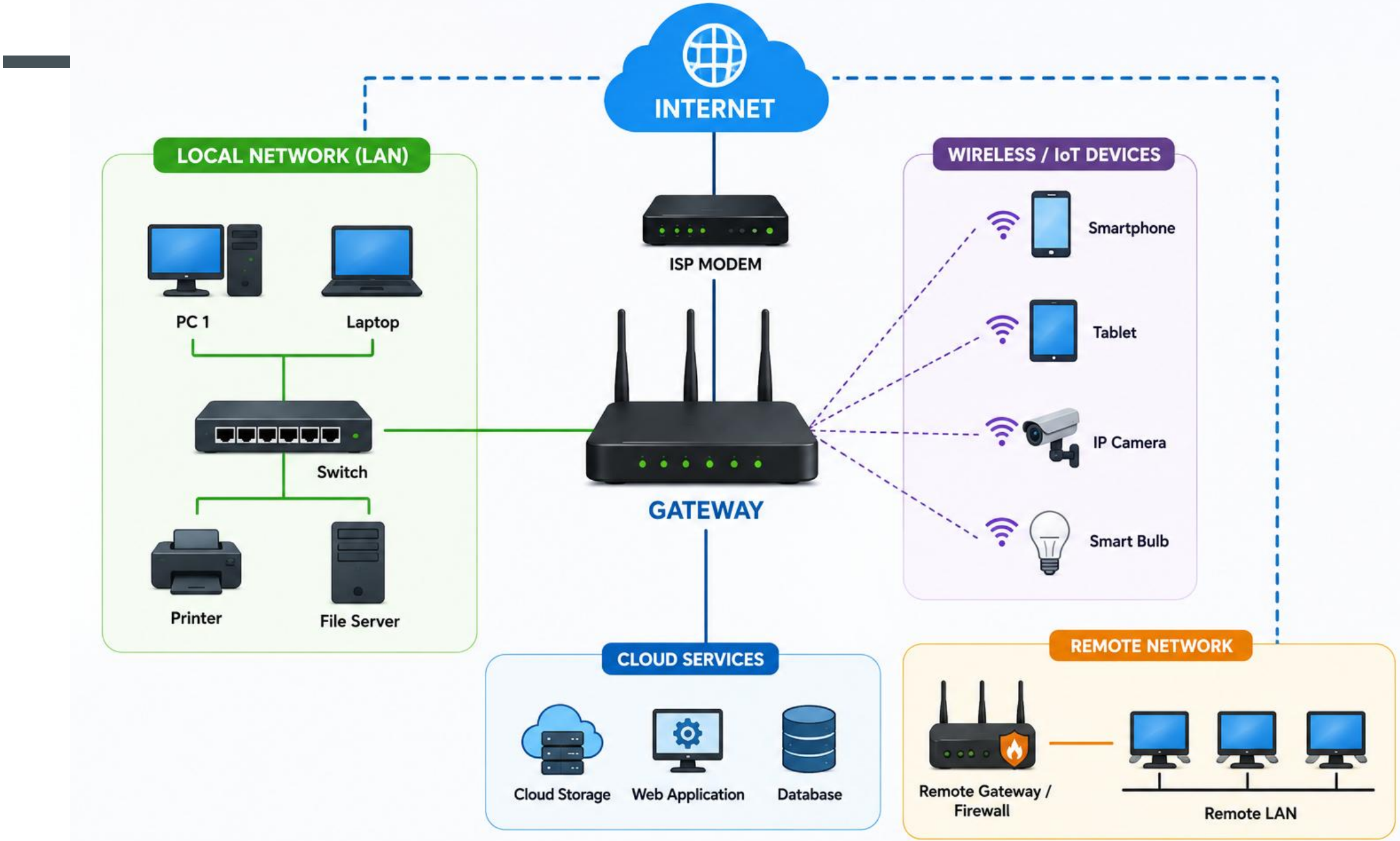
APPLICATIONS

- Connecting LAN to Internet
- Connecting different software or devices
- IoT devices connecting to cloud services
- Enterprise networks & data centers

In Simple Words

Gateway acts as a **bridge + translator** that allows different networks and devices to understand and communicate.




INTERNET
ISP MODEM
LOCAL NETWORK (LAN)
PC 1
Laptop
Switch
Printer
File Server
GATEWAY
WIRELESS / IoT DEVICES
Smartphone
Tablet
IP Camera
Smart Bulb
CLOUD SERVICES
Cloud Storage
Web Application
Database
REMOTE NETWORK
Remote Gateway / Firewall
Remote LAN